

Yan Huang

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Research Interests	Numerical Methods for Kinetic Equations, Gradient Flows, Optimal Transport, Scientific Machine Learning.	
Education	University of Minnesota Twin Cities <i>Ph.D. in Mathematics</i>	Minneapolis, MN, USA Aug. 2023 -
	Southern University of Science and Technology <i>B.S. in Mathematics and Applied Mathematics</i> Honors Program in Mathematics	Shenzhen, China Aug. 2019 - Jun. 2023
Preprints	[3] Y. Huang and L. Wang. Asymptotic-preserving deterministic particle methods for collisional plasma models.	
Publications	[2] Y. Huang and L. Wang. JKO for Landau: a variational particle method for homogeneous Landau equation, IMA Journal of Numerical Analysis, 2026.	
	[1] Y. Huang and L. Wang. A score-based particle method for homogeneous Landau equation, Journal of Computational Physics, 2025, p.114053.	
Presentations	Talks: • SIAM Student Seminar of Mathematical Applications & Computations, UMN, Oct. 27, 2025. • Research Seminar, SUSTech, Oct. 17, 2024.	
	Lightning Talks & Posters: • Learning Models from Data for Multi-Fidelity Fusion Plasma Physics, IPAM, Apr. 15, 2026. • Summer School in Optimal Transport and Applications, UCSB, Jul. 22, 2025. • Statistical and Computational Challenges in Probabilistic Scientific Machine Learning, IMSI, Jun. 10, 2025. • *Rivière-Fabes Symposium on Analysis and PDE, UMN, Apr. 25-27, 2025. • *Workshop on Kinetic Theory and Fluids, UW-Madison, Mar. 28-30, 2025. * Poster only.	
Conference Attending	• John Tukey Summer Graduate School: Mathematics of Generative Models, SLMATH, Jun. 22-Jul. 3, 2026. • Deep Learning meets PDE Workshop, UMN, Apr. 27-29, 2026 • SIAM Conference on Uncertainty Quantification (UQ26), Minneapolis, Mar. 22-25, 2026. • Kinetic Theory: Novel Statistical, Stochastic and Analytical Methods, SLMATH, Aug. 21-29, 2025. • Micro-workshop: Data-driven approaches for physical models, UMN, May. 19, 2025. • PDEs of Incompressible Fluid Flows, Computer Assisted Proofs and Neural Networks, UMN, Apr. 12, 2024. • Computational Challenge and Optimization in Kinetic Plasma Physics, IMSI, Feb. 19-22, 2024.	

- New Trends in Kinetic and Optimal Transport, UMN, Oct. 25-27, 2023.

Services

Journal Reviewer

- Journal of Computational Physics.

Graduate Student Services

- Mentor for Directed Reading Program, UMN, Fall 2024.

Teaching

School of Mathematics, UMN

- Teaching Assistant, MATH 1051 Pre-Calculus, Spring 2026.
- Teaching Assistant, MATH 1571H Honors Calculus I, Fall 2025.
- Teaching Assistant, MATH 1272 Calculus II, Spring 2024, Spring 2025.
- Teaching Assistant, MATH 1271 Calculus I, Fall 2023.
- Grader: MATH 5485 Introduction to Numerical Methods I, MATH 8441 Numerical Analysis and Scientific Computing, MATH 8401 & 8402 Mathematical Modeling and Methods of Applied Mathematics, MATH 8651 & 8652 Theory of Probability including Measure Theory.

Awards

- SUSTech Excellent Undergraduate Thesis.
- SUSTech Second Class of the Merit Student Scholarship, 2021 & 2022.
- First Class Award of the 13th National College Students Mathematics Competition.

Skills

Programming: Python, PyTorch, Matlab, Java.

Language: Chinese (native), English (professional).